1) Intelligent agents in smart home automation

A smart home automation system controls light timing, temperature, and security based on sensor input, users preferences, and environmental conditions.

Types of intelligent agents

⇒ Simple reflex agent: Acts using current sensor input. Example: switches on lights when motion

⇒ Model Based agent: Maintains an internal model of the environment. Eg: remembers room temperature

⇒ Goal-Based Agent: selects actions to achieve a specific goal; Eg: Maintains a comfortable temperature

⇒ Utility Based agent: chooses actions that Maximize comfort, energy efficiency and security Eg: selects the most energy efficient temperature setting.

Best approach: A hybrid of model, based goal-based, and utility-based agents is most effective. It understands the environment, achieves users goals, and balances comfort, energy saving.

Conclusion: Intelligent agent make smart homes more comfortable efficient, safe and automated.

2) A robot in an unknown maze must find the exit without prior knowledge of layout uniformed search explores possible paths without using heuristic information.

uniform cost search (ucs)

* Expands the node with lowest total path cost
* uses a priority queue.
* Finds the optimal solution when steps costs are positive.
* Disadvantage: may consume more time and memory

Iterative deeping search (IDS)

* performs depth-limited search, repeatly, increasing the depth limit each time.
* Requires less memory than UCS.
* Find the shortest path when all steps have equal cost.
* Disadvantage: repeatly explores nodes, causing extra work.

UCS	IDS
* Lowest cost node first. * Priority queue * yes, with positive cost. * High * Different path cost	* Increasing depth limits * stacks/ recursion * for equal step costs * low * Equal cost more path.

A model based agent is suitable because the robot can maintain a map of explored path and obstacles. A goal based agent helps it reach the exit.

Best combination:

A model based goal-oriented agent with UCS is best when maze paths have different costs.

If all moves have equal cost and memory is limited, IDS is suitable alternative.

conclusion:

the agent should update its map while exploring and choose a search strategy based on path cost, memory limits and the need for an optimal solution.